

■ Orderless

`Orderless` is an attribute that can be assigned to a symbol f to indicate that the elements e_i in expressions of the form $f[e_1, e_2, \dots]$ should automatically be sorted into canonical order. This property is accounted for in pattern matching.

The `Orderless` attribute for a function corresponds to mathematical property of commutativity. ■ For an object that represents a matrix or a tensor, the `Orderless` attribute represents symmetry among indices. ■ Functions like `Plus` and `Times` are `Orderless`. ■ In matching patterns with `Orderless` functions, all possible orders of arguments are tried. ■ The `Orderless` attribute must be assigned before defining any values for an `Orderless` function. ■ See page 214. ■ See also: `Sort`, `Flat`, `OneIdentity`.